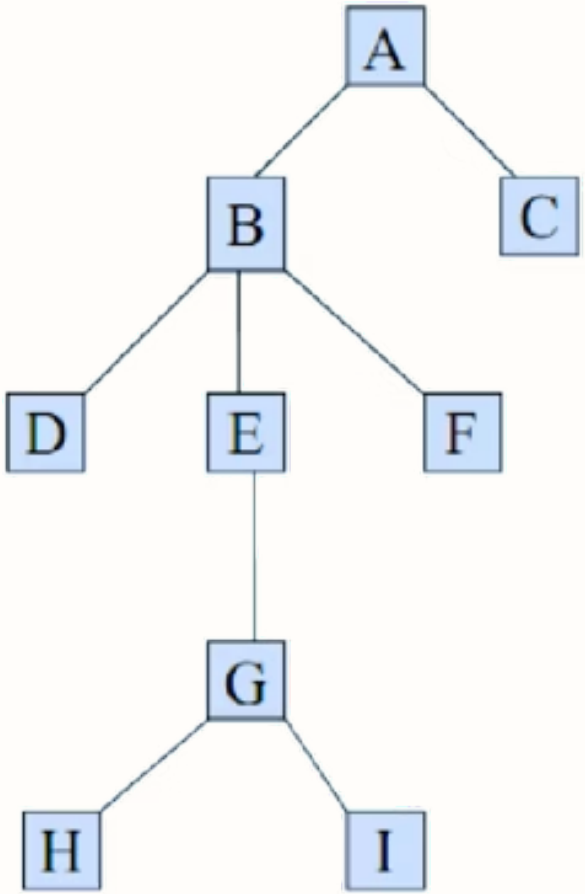
Ağaç Veri Yapısı Soruları

AS by Aylin Sabancı

Made with GAMMA

Soru 1



Düğüm Sayısı:

Kök Düğüm:

Yapraklar:

H'nin Ataları:

B'nin Torunları:

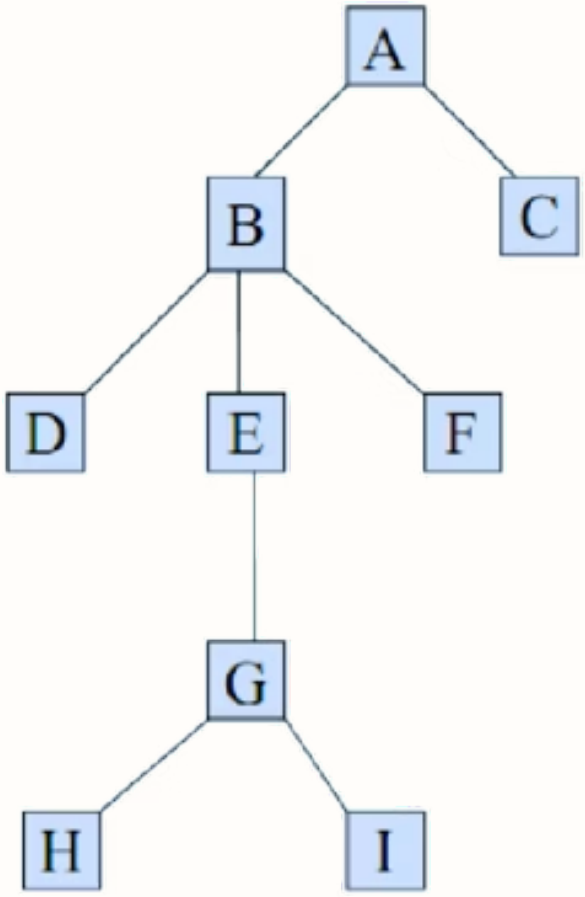
E'nin Kardeşleri:

Seviye sayısı:

Yükseklik:

Made with GAMMA

Soru 1 - Cevap



Düğüm Sayısı: 9

Kök Düğüm: A

Yapraklar: C, D, F, H, I

H'nin Ataları: G, E, B, A

B'nin Torunları: E, G, H, I

E'nin Kardeşleri: D, F

Seviye sayısı: 5

Yükseklik: 4

Made with GAMMA

Soru 2

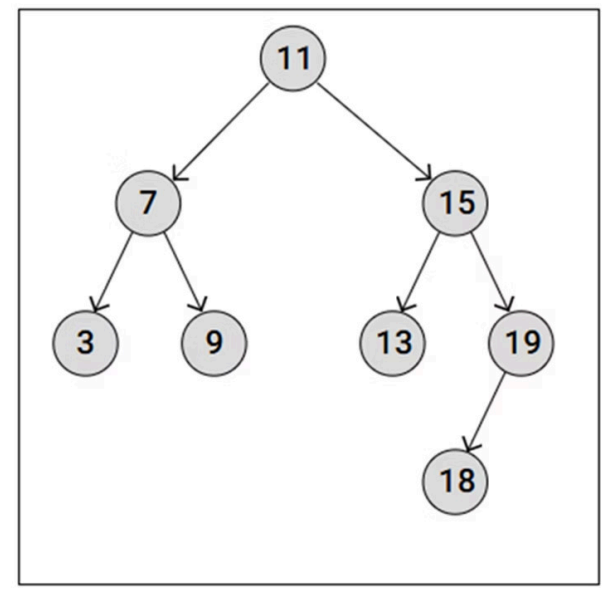
İkili Ağaç Veri Yapısı Türleri

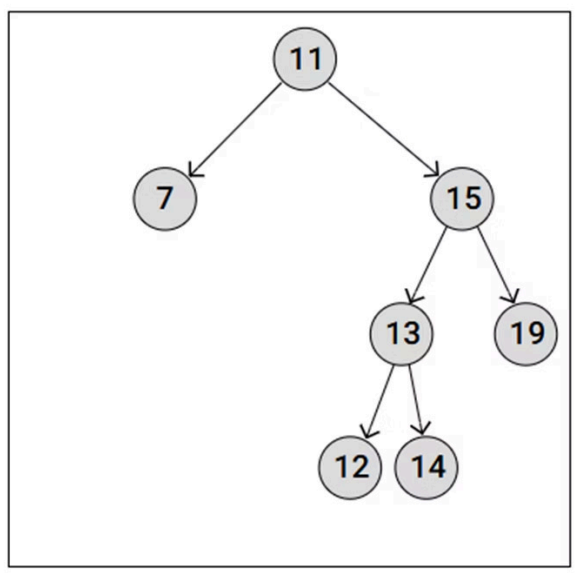
Full Binary Tree

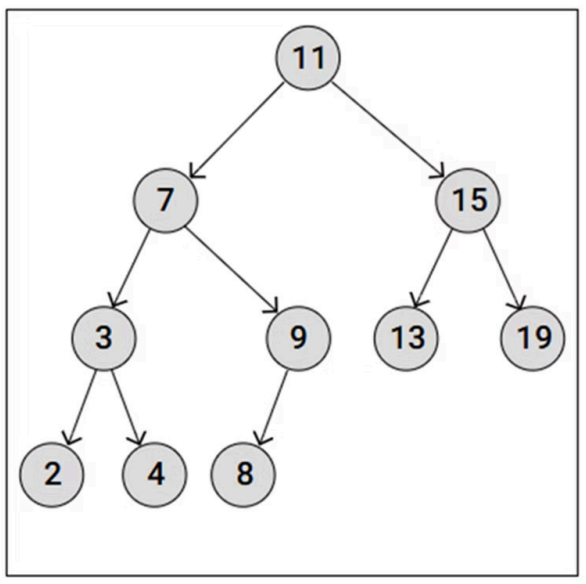
Complete Binary Tree

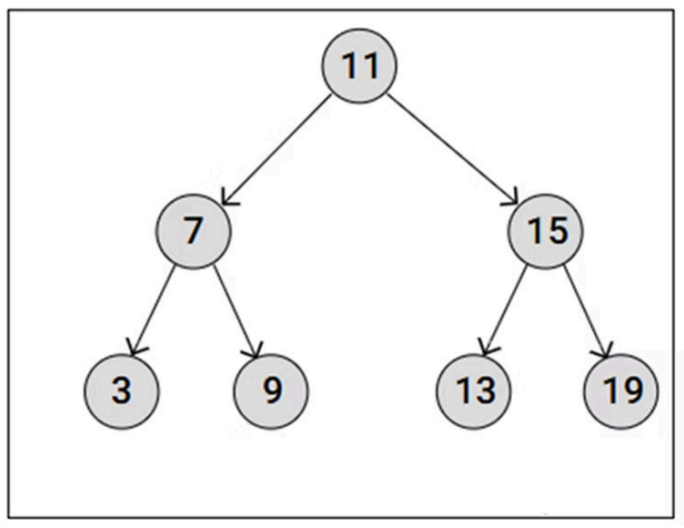
Balanced Binary Tree

Perfect Binary Tree



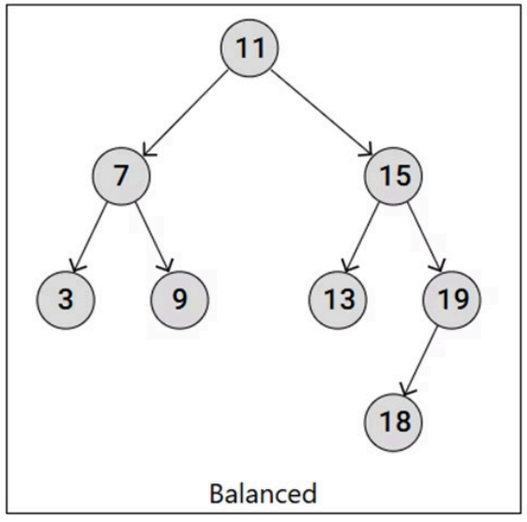




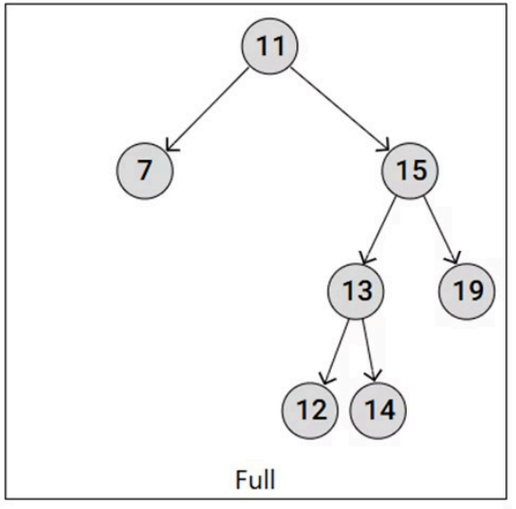


Made with GAMMA

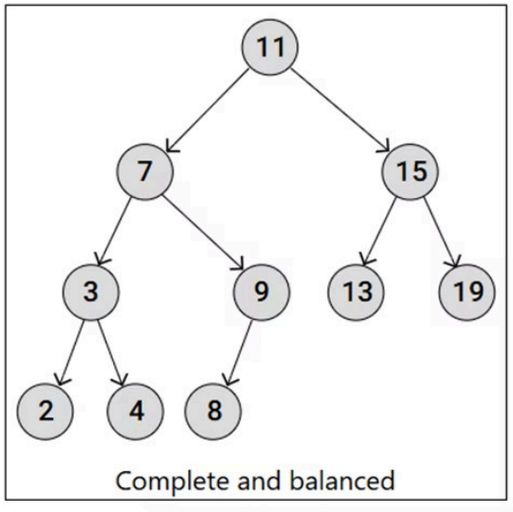
Soru 2 - Cevap



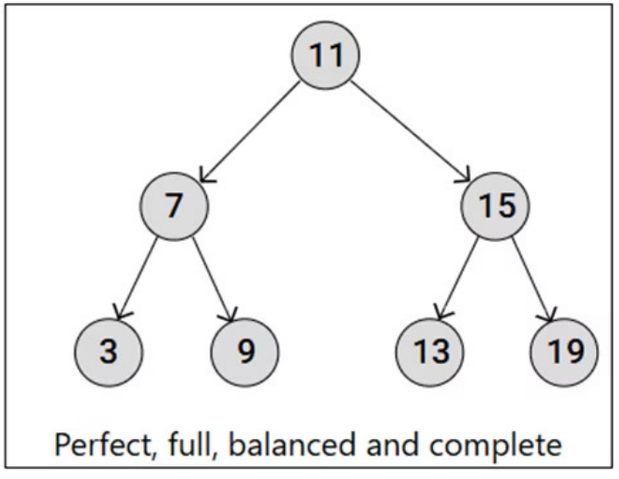
Balanced



Full



Complete and balanced

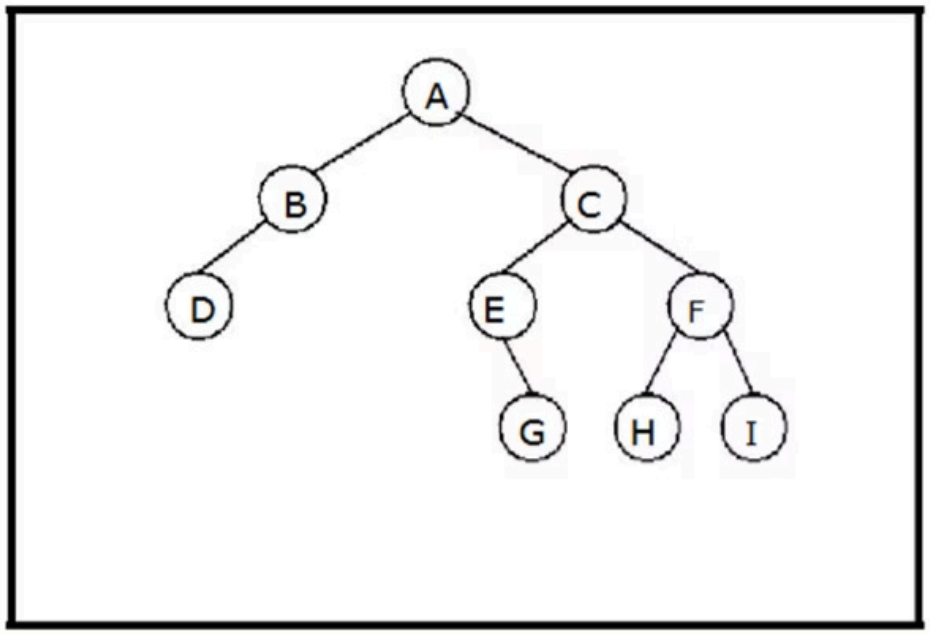


Perfect, full, balanced and complete

Made with GAMMA

Soru 3

Binary Tree Traversal (İkili Ağaçlar Üzerinde Dolaşma)



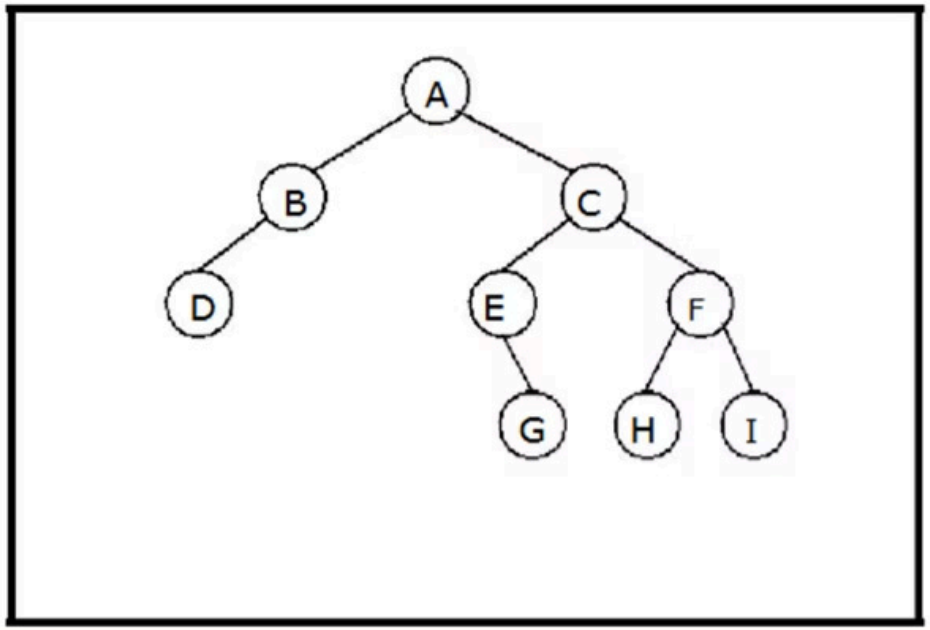
Preorder:

Inorder:

Postorder:

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Soru 3 - Cevap



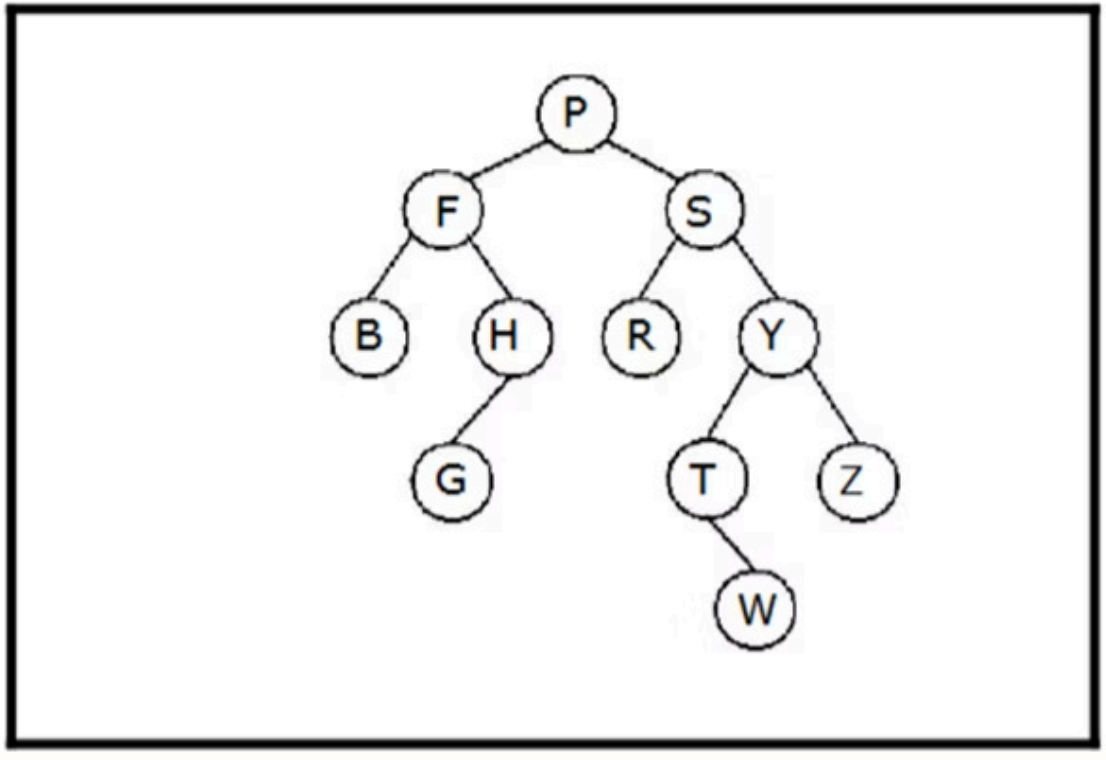
Preorder: A, B, D, C, E, G, F, H, I

Inorder: D, B, A, E, G, C, H, F, I

Postorder: D, B, G, E, H, I, F, C, A

Made with GAMMA

Soru 4



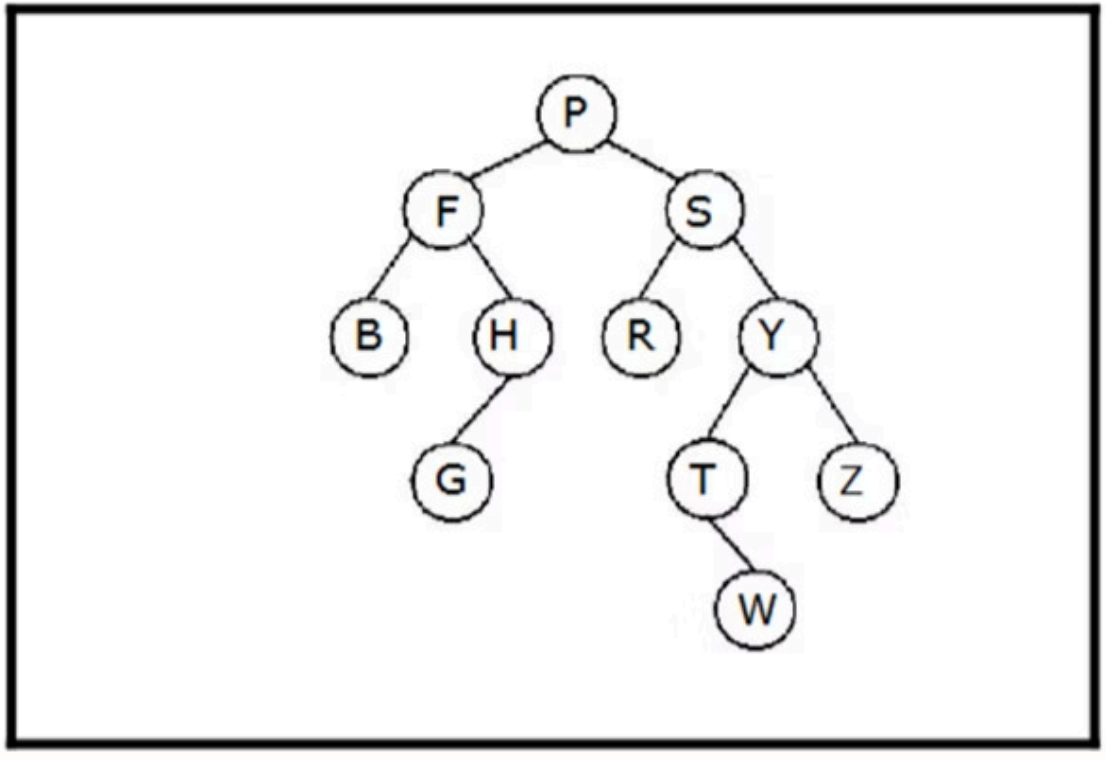
Preorder:

Inorder:

Postorder:

Made with GAMMA

Soru 4 - Cevap



Preorder: P, F, B, H, G, S, R, Y, T, W, Z

Inorder: B, F, G, H, P, R, S, T, W, Y, Z

Postorder: B, G, H, F, R, W, T, Z, Y, S, P

Made with GAMMA

Soru 5

İkili Arama Ağaçları (Binary Search Trees)

- BST'de parent ve child düğümleri arasında büyüklük küçüklük ilişkisi var

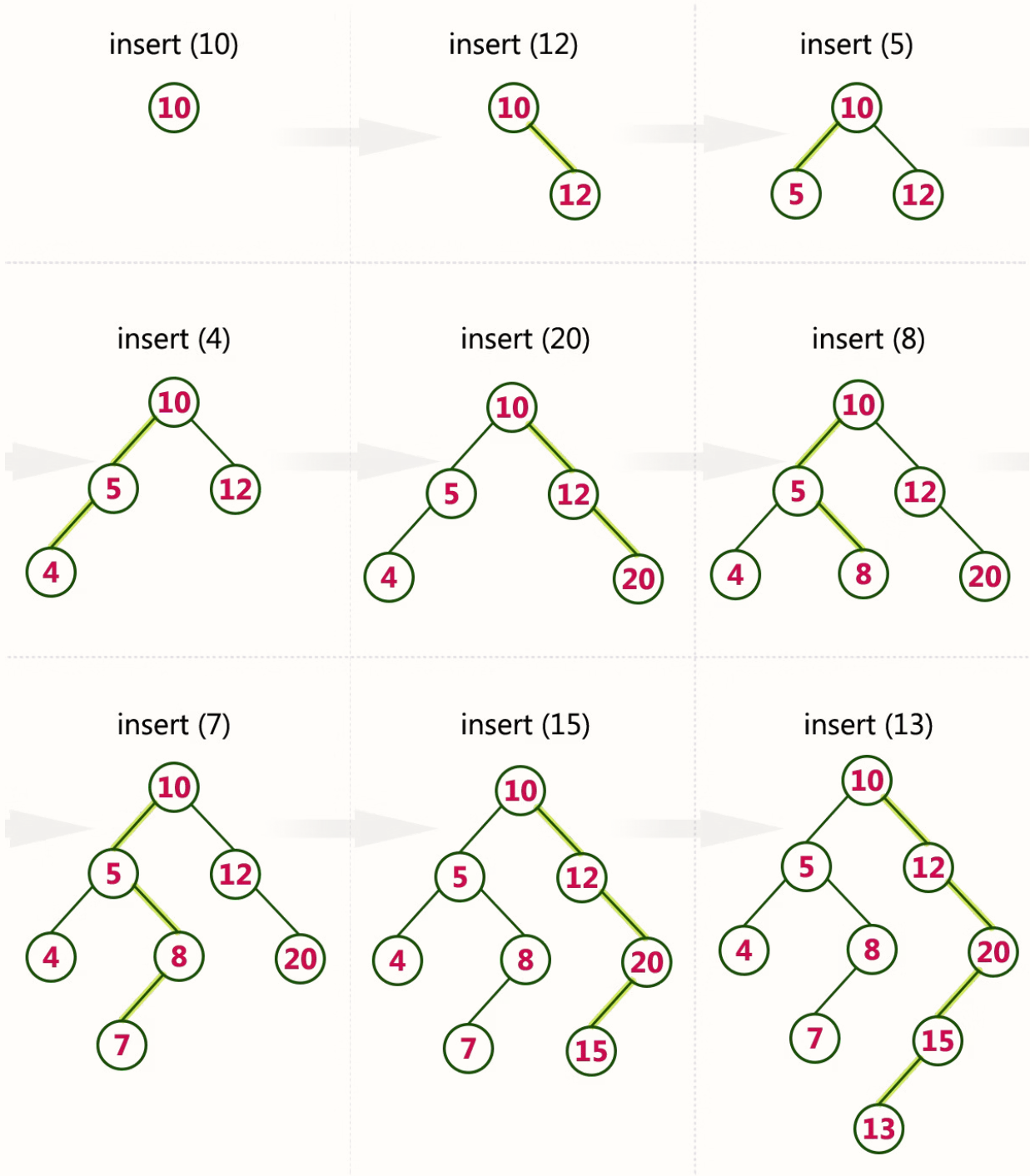
Her düğüm `v` için BST koşulu:

- `left(v)` alt ağacındaki tüm düğümler `< key(v)`
- `right(v)` alt ağacındaki tüm düğümler `> key(v)`

[10, 12, 5, 4, 20, 8, 7, 15, 13] bu diziyi kullanarak BST oluşturma

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Soru 5 - Cevap



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Soru 6

AVL Ağacı

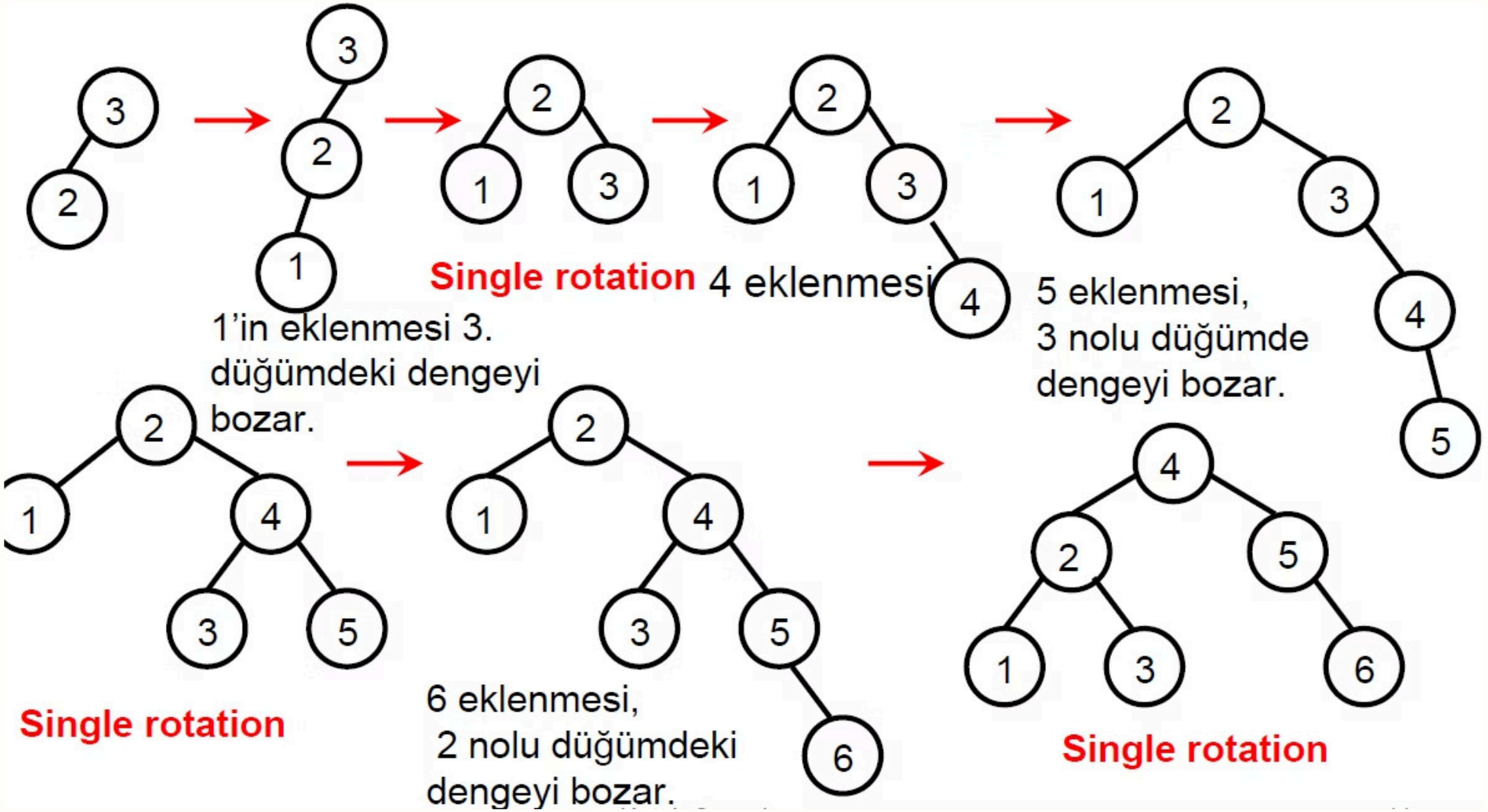
AVL ağacı → Denge koşulu olan ikili arama ağacı

Her düğüm `v` için denge faktörü (DF) → sol ağaç yüksekliği - sağ ağaç yüksekliği [-1, 0, 1]

Sırayla 3, 2, 1, 4, 5, 6 verilerini AVL ağacına ekleyelim

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Soru 6 - Cevap



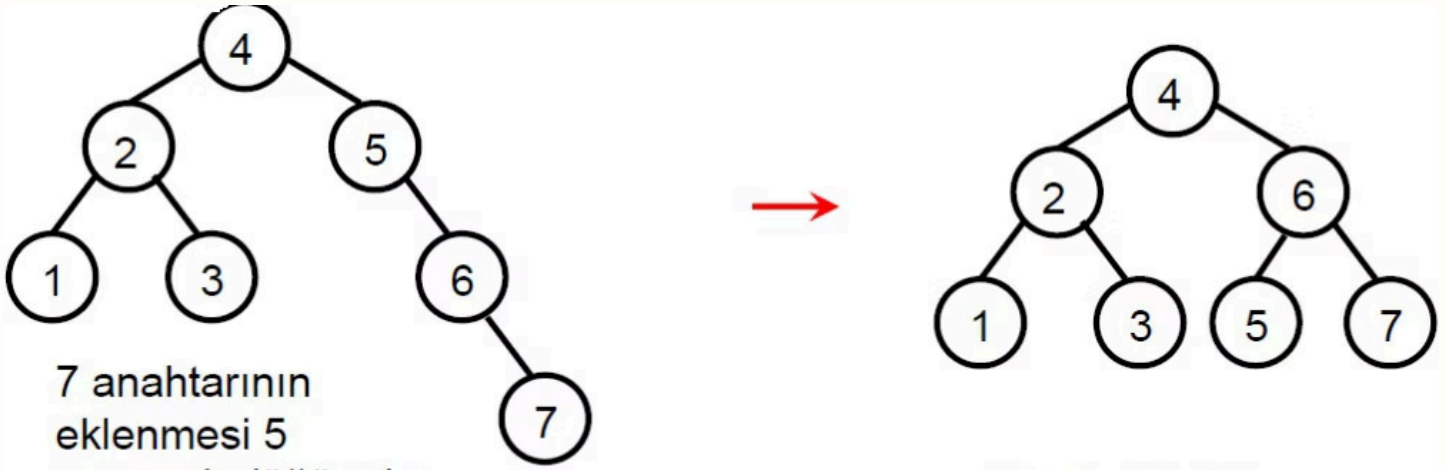
Made with GAMMA

Soru 7

Sırayla 7, 16, 15, 14, 13, 12, 11, 10, 8, 9 verilerini AVL ağacına ekleyelim

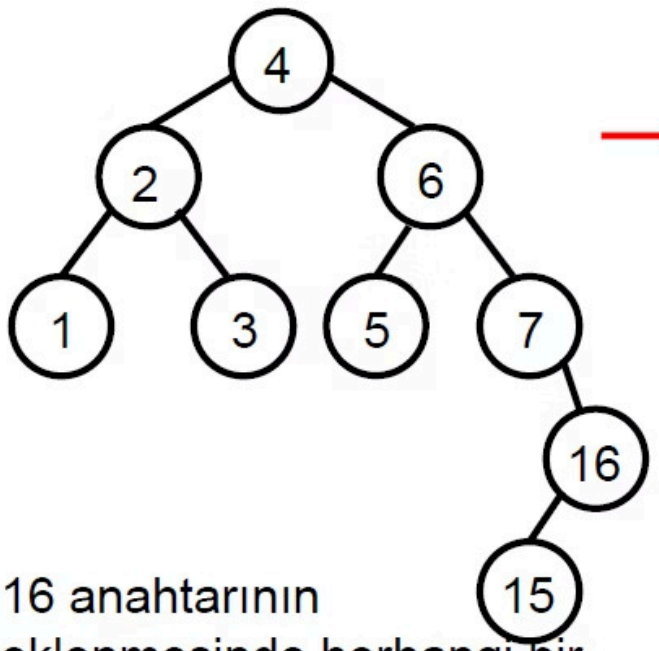
Made with GAMMA

Soru 7 - Cevap

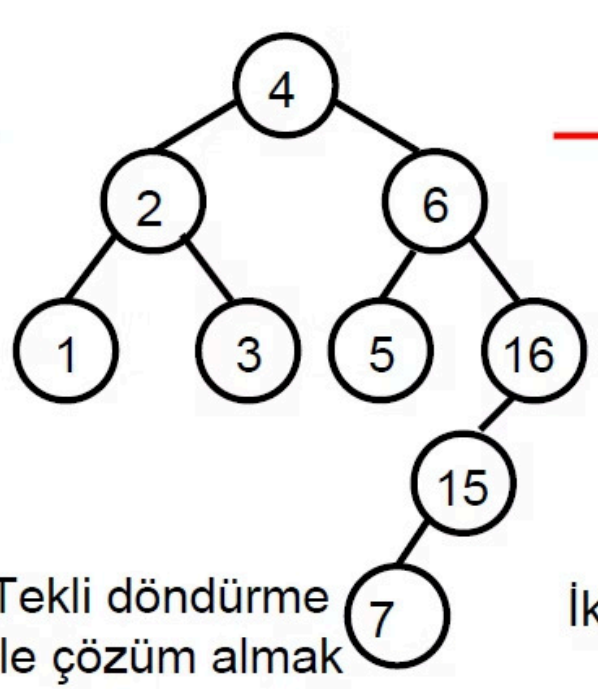


7 anahtarının eklenmesi 5 numaralı düğümde dengesizliğe neden olur.

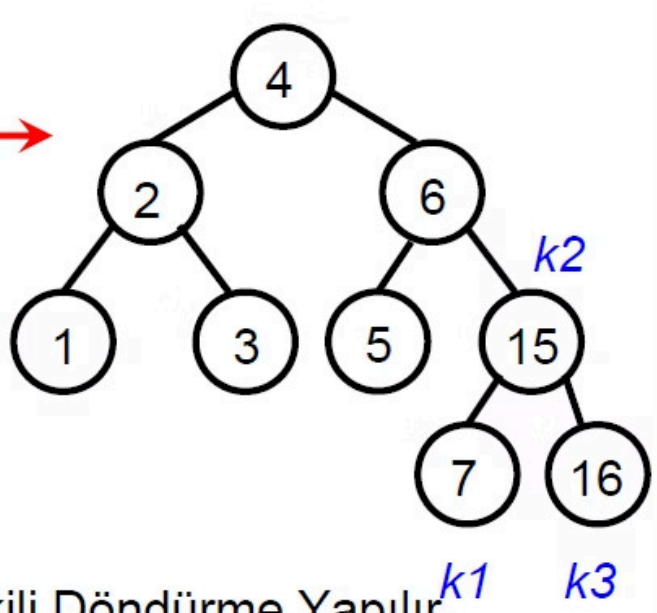
Single rotation



16 anahtarının eklenmesinde herhangi bir problem yok iken 15 anahrarının eklenmesiyle 7 numaralı düğümde dengesizlik meydana gelir.



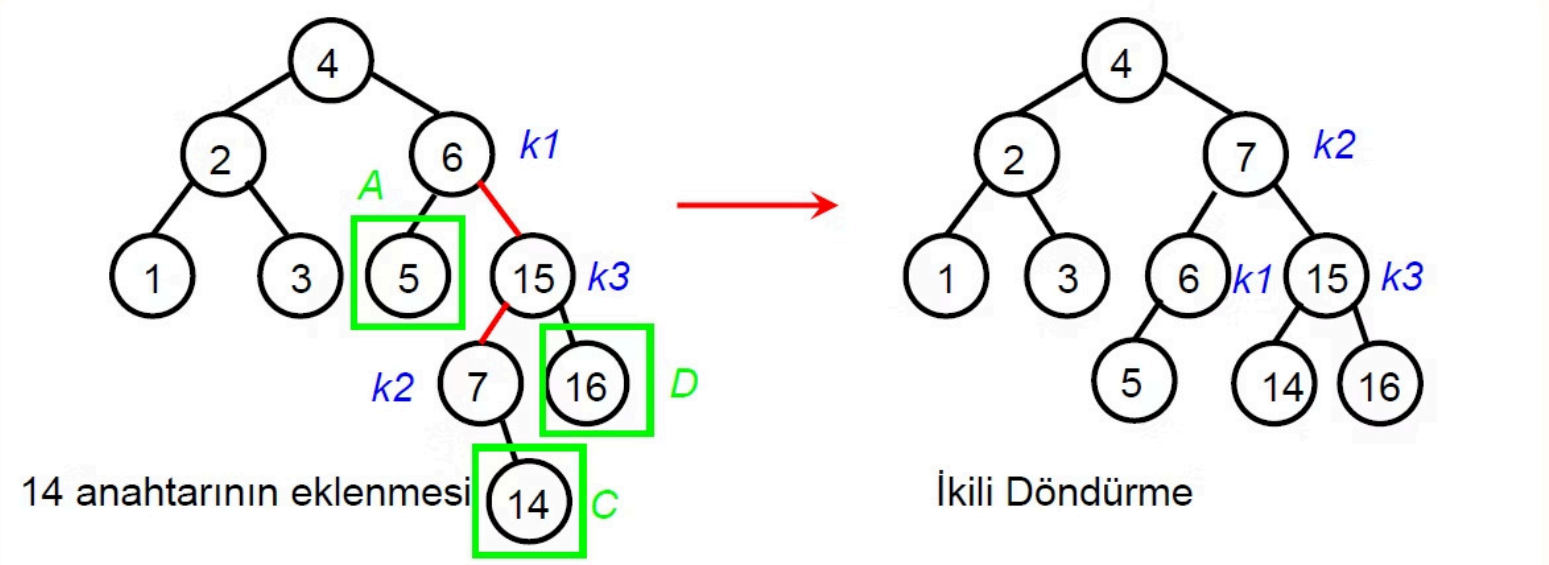
Tekli döndürme ile çözüm almak mümkün değildir. Hala dengesiz durum devam etmektedir.

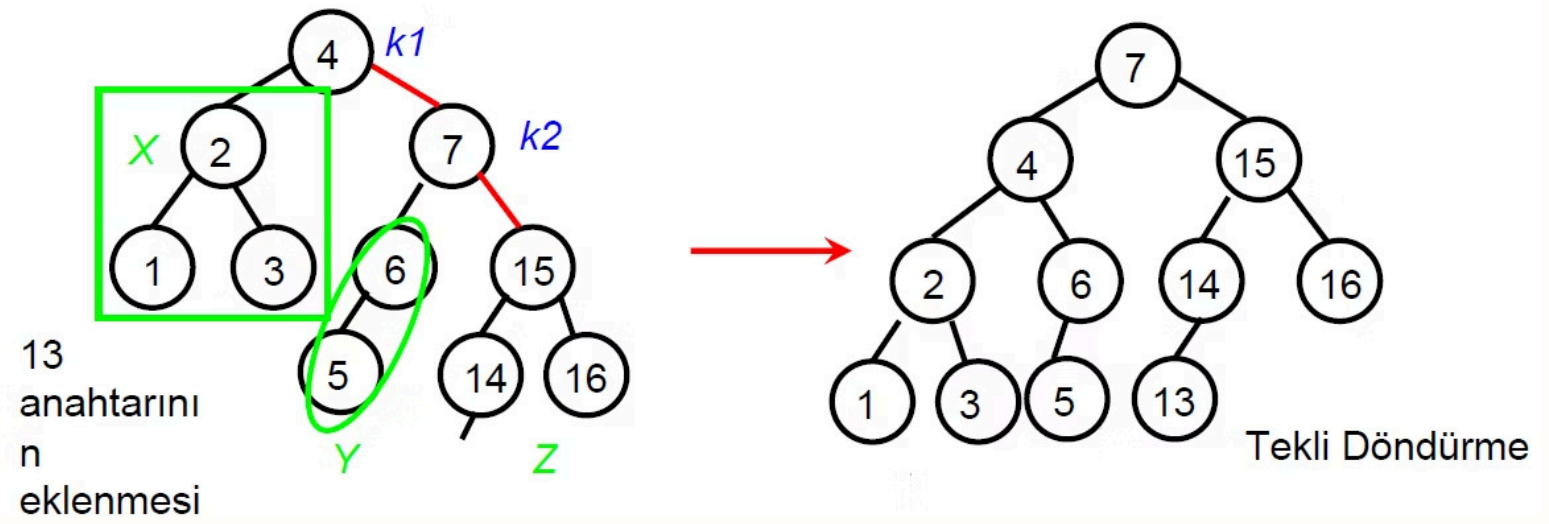


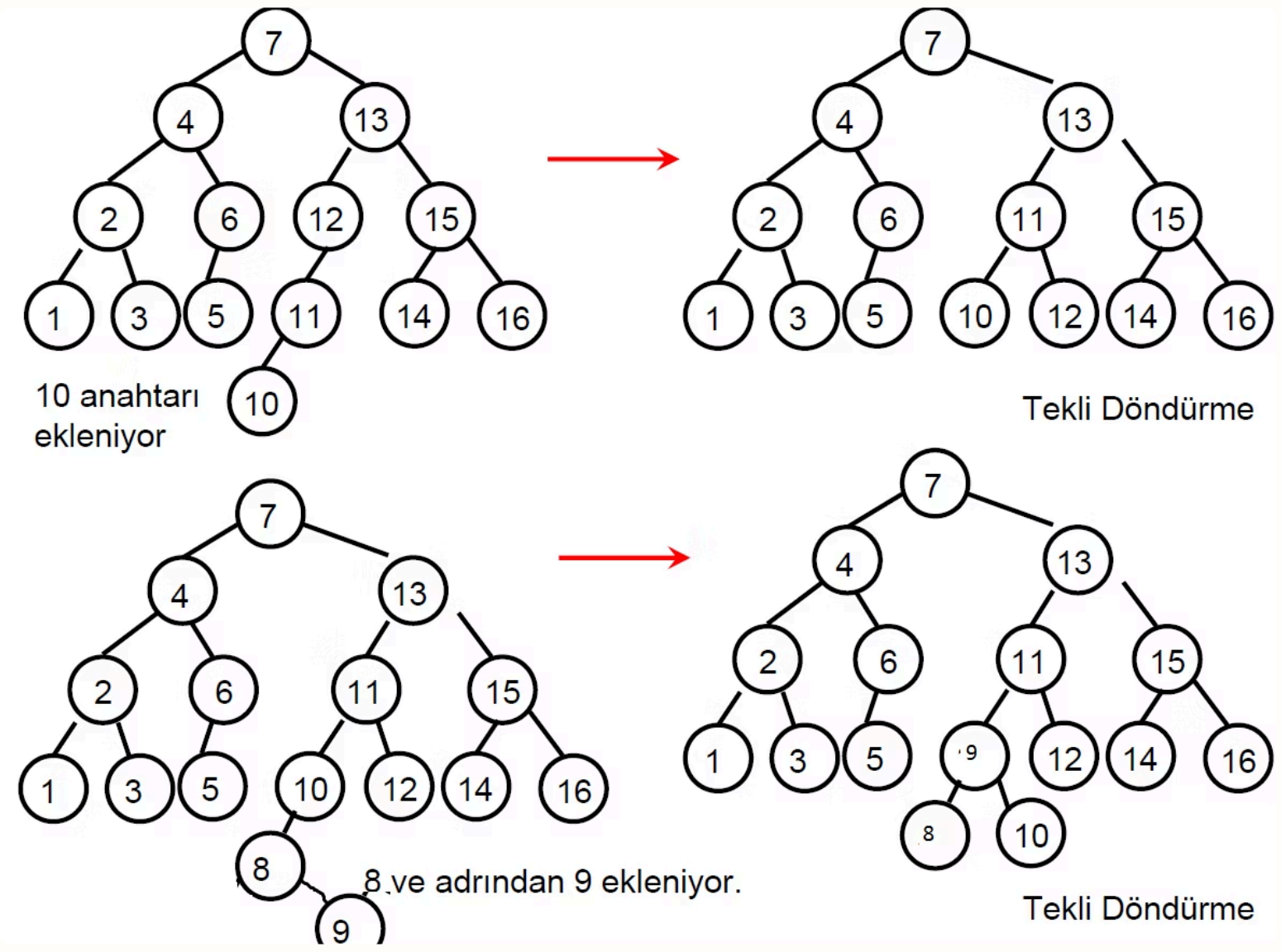
İkili Döndürme Yapılır

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Soru 7 - Cevap







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